

Aviation Safety System — MVP Technical & Business Document (V 1.2)

Project: SkyDeck Aviation Reporting & Document System

Purpose: Stakeholder MVP Demonstration

Status: Pre-Production MVP

Audience: Technical Review + Stakeholders

1. PROJECT PURPOSE

Deliver a secure, scalable aviation reporting & document platform MVP to demonstrate business value and secure funding for full production development.

System digitizes:

- Safety reporting
 - Operational forms
 - Aviation manuals
 - Controlled document distribution
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2. MVP vs PRODUCTION SCOPE

MVP (10-day showcase)

Included:

- Authentication & roles
- Multi-device login
- Manual library (4 PDFs)
- Strict server-side watermarking
- One digital form + submission
- Admin panel (upload/view)
- Audit logging (basic)
- Secure file access
- Backup strategy with RTO/RPO
- Form versioning foundation

Not included (production):

- Full offline-first engine
- Native mobile wrapper
- Advanced encryption key vault
- Multi-airline production scale
- Push notifications infra
- High-availability clustering

3. USER ROLES

Pilot

- Login multi-device
- View/download manuals
- Fill & submit form
- No edit after submit

Admin

- Upload/remove manuals
- View all submissions
- Access dashboard

Viewer/Manager

- View submissions only

All access enforced server-side only.

4. SECURITY ARCHITECTURE

4.1 Authentication & Session Management

JWT-based auth with server session tracking.

Sessions table (updated)

Each login creates session record.

sessions

- id
- user_id
- device_info
- refresh_token_hash
- created_at
- expires_at
- revoked

Purpose:

- Multi-device control
- Token expiry tracking
- Future remote logout

Access token:

- short lifetime

Refresh token:

- stored hashed in DB
- expires tracked via expires_at

4.2 Strict Server-Side Watermarking (MANDATORY)

All manuals must be watermarked server-side BEFORE delivery.

No client-side watermark option allowed.

Process:

1. User requests manual
2. Server verifies auth & role
3. Server generates watermarked version dynamically
4. Watermark embedded into PDF
5. File delivered to user

Watermark must include:

- Username
- Timestamp
- Unique hash ID
- Confidential notice

Goal:

Prevent document leakage & ensure traceability.

4.3 File Access Control

Files are never public.

Rules:

- Stored outside public web root
- Access only through authenticated API
- Role checked before delivery
- Every download logged

Pilot → manuals only

Admin → manuals + submissions

Viewer → submissions only

4.4 Brute-Force & Session Protection

- Login rate limiting
 - Temporary lock after repeated failures
 - Failed login logging
 - Token expiry enforcement
 - Refresh rotation ready (future)
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5. AUDIT LOGGING (EXPANDED)

System logs must include:

Authentication

- login success/failure
- logout

Data actions

- form submit
- manual upload/delete
- file download/view

Admin actions

- manual management
- submission viewing

Audit log fields:

- user_id
- action
- target
- timestamp
- IP/device (if available)

Logs append-only.

6. FORM VERSIONING (PREPARED)

Even with one form, system must support versioning.

Structure:

- form_templates
- form_versions
- submissions linked to version

Each submission:

- immutable
 - linked to form_version_id
 - reproducible later
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7. DATABASE SCHEMA (MVP)

users

- id
- name
- email
- password_hash
- role

- created_at

sessions (updated)

- id
- user_id
- device_info
- refresh_token_hash
- created_at
- expires_at
- revoked

manuals

- id
- title
- file_path
- uploaded_by
- created_at

form_templates

- id
- name

- created_at

form_versions

- id
- template_id
- version_number
- schema_json
- created_at

submissions

- id
- user_id
- form_version_id
- data_json
- attachment_path
- hash_id
- created_at

audit_logs

- id
- user_id

- action
 - target
 - timestamp
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8. FILE STORAGE STRATEGY

MVP

Local server storage:

- /storage/manuals
- /storage/submissions

Served via backend only.

Production future

- Object storage (S3 compatible)
 - Presigned secure delivery
 - CDN optional
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9. MULTI-DEVICE SESSION SUPPORT

Users may login from:

- iPad
- iPhone
- Desktop

Behavior:

- Multiple sessions allowed
- Each tracked in sessions table
- Expiration enforced
- Future remote revoke ready

10. BACKUP & DISASTER RECOVERY

Backup Scope

Daily automated backups:

- Database
- Uploaded files
- Manuals

Stored:

- Separate backup directory or server

MVP Recovery Targets

RPO (Recovery Point Objective): 24 hours

Maximum acceptable data loss = 1 day

RTO (Recovery Time Objective): 4 hours

System restorable within 4 hours

These are MVP-level targets and will improve in production.

11. COMPLIANCE ALIGNMENT

System concept aligns with aviation SMS principles:

- Confidential safety reporting
- Non-punitive reporting culture
- Traceable records
- Controlled access
- Data integrity

Prepares system for ICAO/EASA-aligned workflows.

12. BUSINESS VALUE

This system enables:

- Paperless safety reporting

- Secure manual distribution
 - Traceable confidential reporting
 - Multi-airline SaaS potential
 - Foundation for aviation compliance tools
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13. MVP DEMO FLOW

1. Pilot login
2. Open manual → server-side watermark visible
3. Fill report
4. Submit
5. Admin login
6. View submission
7. Upload manual
8. Pilot sees new manual

Demonstrates full workflow.

14. MVP SUCCESS CRITERIA

MVP successful if:

- ✓ secure login works

- ✓ sessions tracked with expiry
 - ✓ server-side watermark enforced
 - ✓ manual access controlled
 - ✓ form submission stored
 - ✓ admin sees reports
 - ✓ audit logs active
 - ✓ backup functioning
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15. POST-MVP NEXT STEP

After approval:

- Full offline-first architecture
- Native mobile wrapper
- Advanced encryption key management
- Multi-airline tenancy
- Production DevOps & monitoring
- Notification system